

Unit-3

Production and Related Concepts

Meaning of Production

- • Production refers to the creation of utility.
- • It is the process of transforming inputs into outputs.
- • Objective: To satisfy human wants by producing goods and services.
- • Production includes both goods and services.

Factors of Production

- • Land – Natural resources used in production.
- • Labour – Human effort (physical & mental).
- • Capital – Man-made resources (tools, machines, buildings).
- • Entrepreneurship – Risk-taking and organizing factors.

Production Function

- • It represents the relationship between inputs and outputs.
- • General form: $Q = f(L, K)$ where Q =Output, L =Labour, K =Capital.
- • Short-run: At least one factor is fixed.
- • Long-run: All factors are variable.

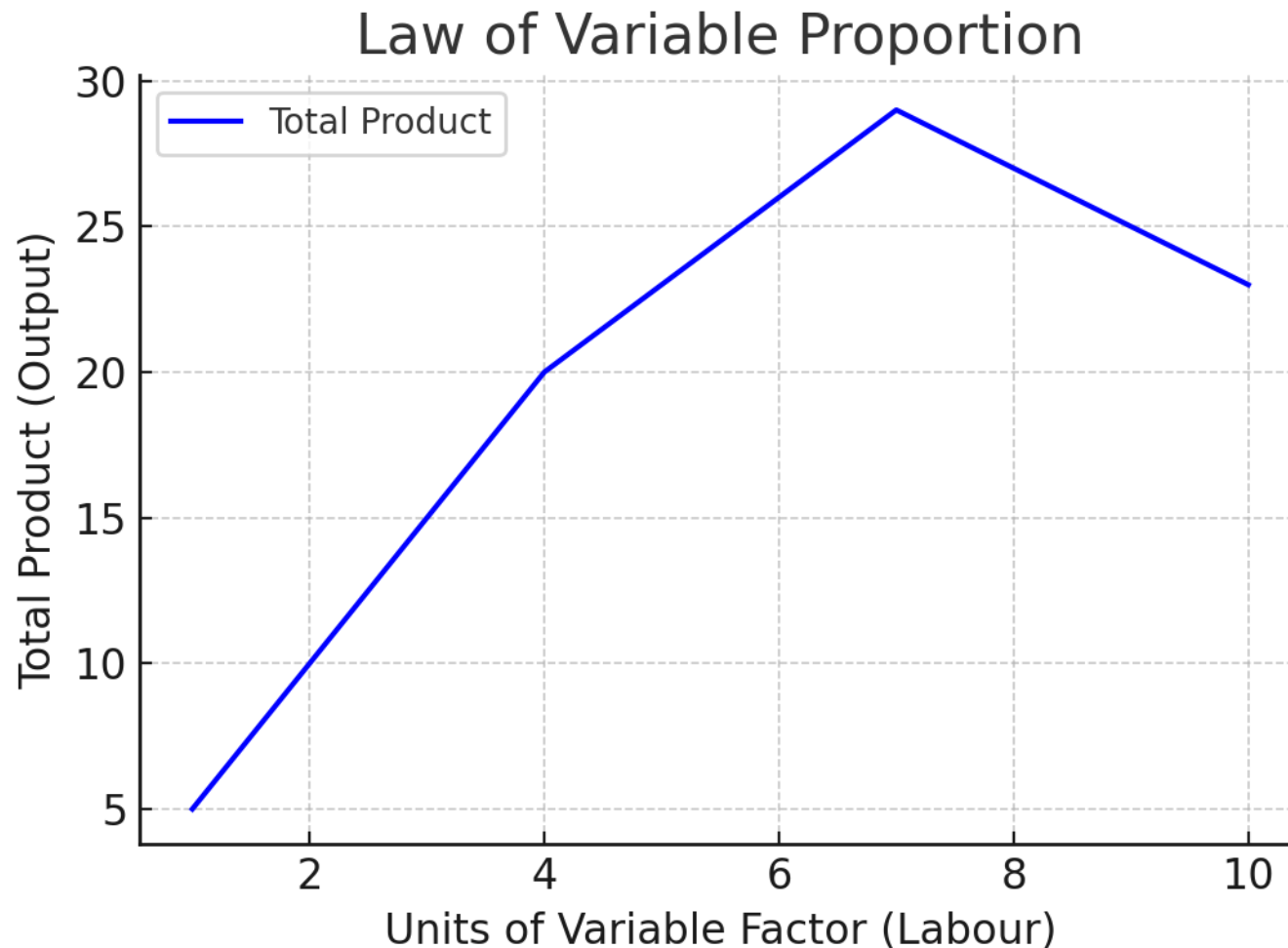
Fixed and Variable Factors

- • Fixed Factors: Do not change with output (e.g., land, buildings).
- • Variable Factors: Change with output (e.g., labour, raw materials).
- • Short-run: Combination of fixed and variable factors.
- • Long-run: All factors become variable.

Law of Variable Proportion

- • Applicable in the short-run when one factor is variable.
- • States that as more units of a variable factor are employed,
 - total product increases at increasing rate, then diminishing rate,
 - and eventually falls.
- • Stages: Increasing Returns → Diminishing Returns → Negative Returns.

Law of Variable Proportion (Diagram)



Law of Returns to Scale

- • Applicable in the long-run when all factors are variable.
- • Increasing Returns to Scale: Output increases more than proportionately.
- • Constant Returns to Scale: Output increases proportionately.
- • Decreasing Returns to Scale: Output increases less than proportionately.

Law of Returns to Scale (Diagram)

